

10 A vector $\mathbf{v}$ has magnitude 1 unit. The angle between $\mathbf{v}$ and the positive z -axis is 60° , and $\mathbf{v}$ is parallel to the plane $x - 2y = 0$.

Given that $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, where a , b and c are all positive, find $\mathbf{v}$. **[7]**